₹4 CRORE

What Indians will have to pay in the US for CAR-T therapy, which is currently unavailable in India

THE CHANGE-MAKERS



IMMUNEEL THERAPEUTICS

- Founded in 2019 by Kiran Mazumdar-Shaw, Dr Kush Parmar and Dr Siddhartha Mukherjee, Immuneel has filed India's first CAR-T application for approval
- The company has exclusive rights to varnimcabtagene autoleucel (Var-Cel), a CAR-T therapy approved in Spain from Hospital Clínic Barcelona for further clinical trials, manufacturing and commercialisation in India via technology transfer
- Results presented by Immuneel at global meetings in the US and Europe this year indicate 77 per cent positive response at three months (including most with complete response) in patients treated with Var-cel in India



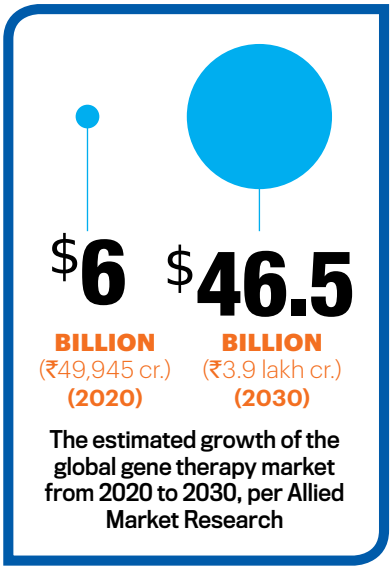
LAURUS LABS

- In June 2023, Laurus Labs signed an agreement with IIT Kanpur to bring new gene therapy products to the market
- Under the agreement, Laurus will in-license a small number of gene therapy products and offer research grants to advance them through pre-clinical development
- It will also establish a Good Manufacturing Practice (GMP)-compliant facility at IIT Kanpur's Techno Park facility
- It also has a 40 per cent stake in ImmunoACT, which is in the second phase of trials for a CAR-T product for blood cancer

tering a compound annual growth rate of 22.8 per cent.

WHY IT IS A GAME CHANGER

Gene therapy works to treat or prevent a disease by either adding new copies of a broken gene, or altering a gene or inserting a missing one. Its genesis goes back decades, to the 1960s and early 1970s, when researchers were able to isolate certain genes from the DNA. By 1990, the medical world was already boasting its first success story in Ashanti de Silva, whose severe combined immunodeficiency disease (SCID), caused by insufficient levels of the enzyme adenosine deaminase (ADA), was treated by being injected with a healthy version of the gene that produces ADA. Since then, gene therapy as an area of study has attracted wide interest from researchers and the pharma industry alike due to its breakthrough potential for the treatment of disorders like haemophilia, sickle cell disease, and even leukaemia.



The estimated growth of the global gene therapy market from 2020 to 2030, per Allied Market Research

WHAT INDIA NEEDS TO MASTER

In gene therapy, India's biggest challenge will be affordability. Research from industry analyst GlobalData indicates that there are currently only four gene therapies being developed in India: two blood cancer-targeting CAR-T therapies by Immuneel Therapeutics and the Laurus Labs-backed ImmunoACT; Stempeutics Research's Stempeucel for osteoarthritis, diabetic foot ulcers and Crohn's disease; and Cancer institute WIA's cellular immunotherapy for HPV-associated cervical cancer and ovarian cancer. CAR-T, which involves collecting immune cells (T-Cells) from the body and modifying them so that they can recognise and fight cancer, has been in market in the US and other countries since 2017, but it's not available in India yet. If a patient were to avail of the therapy in the US, they would have to spend nearly Rs 4 crore. Further, even if CAR-T therapy does get indigenous approval, the treatment is expected to cost Rs 20-30 lakh. Thus, in time, steps must be taken to make gene therapy affordable, as well as to encourage more private players to invest in the field. Also, to compete globally, India must speed up production while also maintaining strict ethical practices. Insurance coverage of gene therapy products can help, and logistics, such as cold storage links, will have to be beefed up too.

India stands at the cusp of a new era. The challenge now is to create not just an enabling but also a progressive framework for this fast-evolving field. ■